

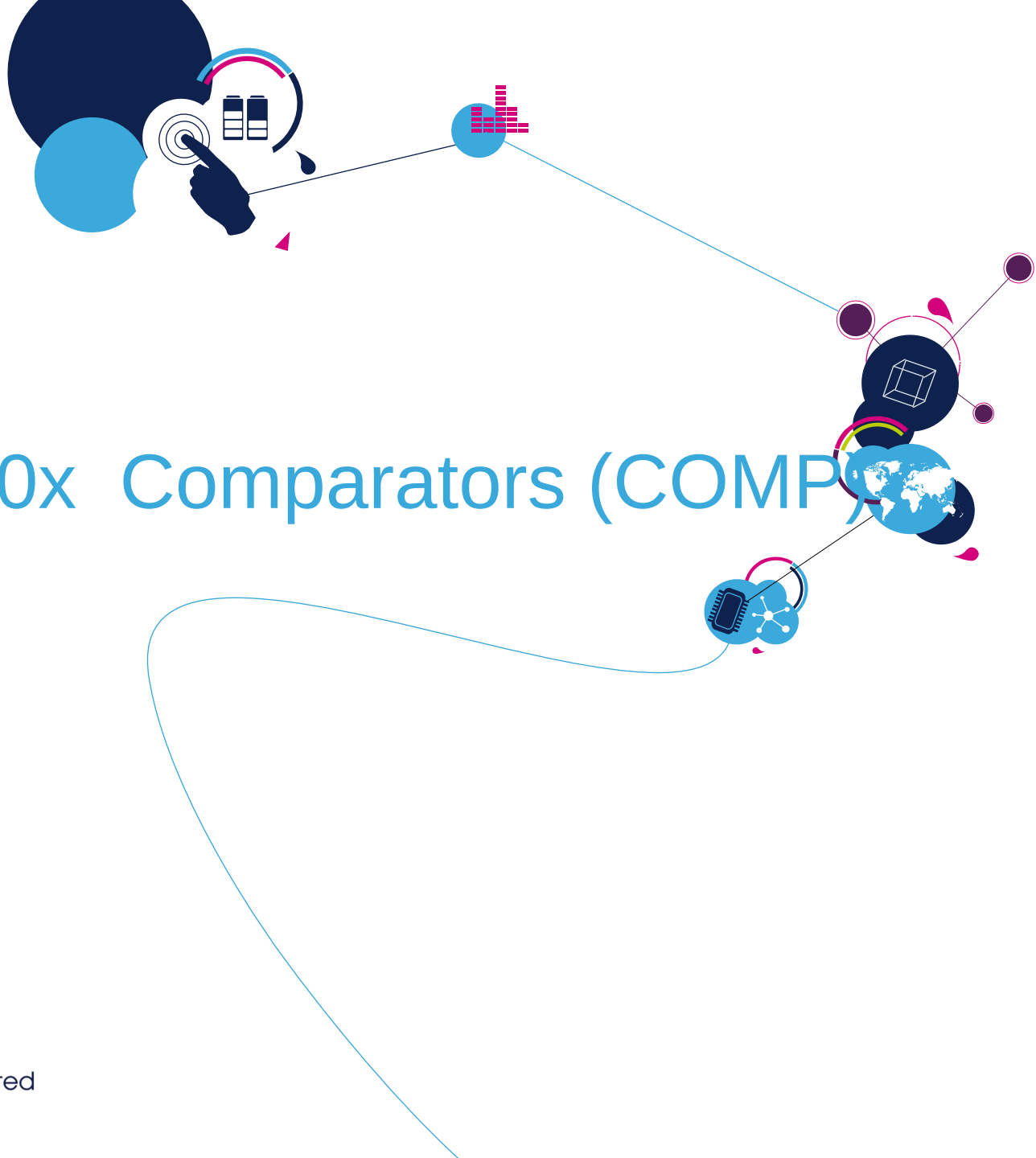


STM32F3 Technical Training

For reference only

Refer to the latest documents for details





STM32F30x Comparators (COMP)

COMP features (1/2)

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- 7 comparator pairs COMP_x, $x = 1..7$
 - Rail-to-rail inputs
 - Programmable speed / consumption: 4 modes
 - Programmable hysteresis: 4 levels
 - Inputs and outputs available externally - can be used as a standalone device without MCU interaction
 - Comparator pairs can be combined into a window comparators
 - Multiple choices for output redirection
 - Comparator blanking – The blanking time period is defined by TIM OC – multiple timer OC events available
 - to avoid reaction of the regulation loop on the current spikes at the beginning of the PWM period caused by the recovery current in power switches
- Can be used for:
 - Exiting low power modes
 - Signal conditioning
 - Cycle-by-cycle current control with blanking (w/ DAC and TIM)

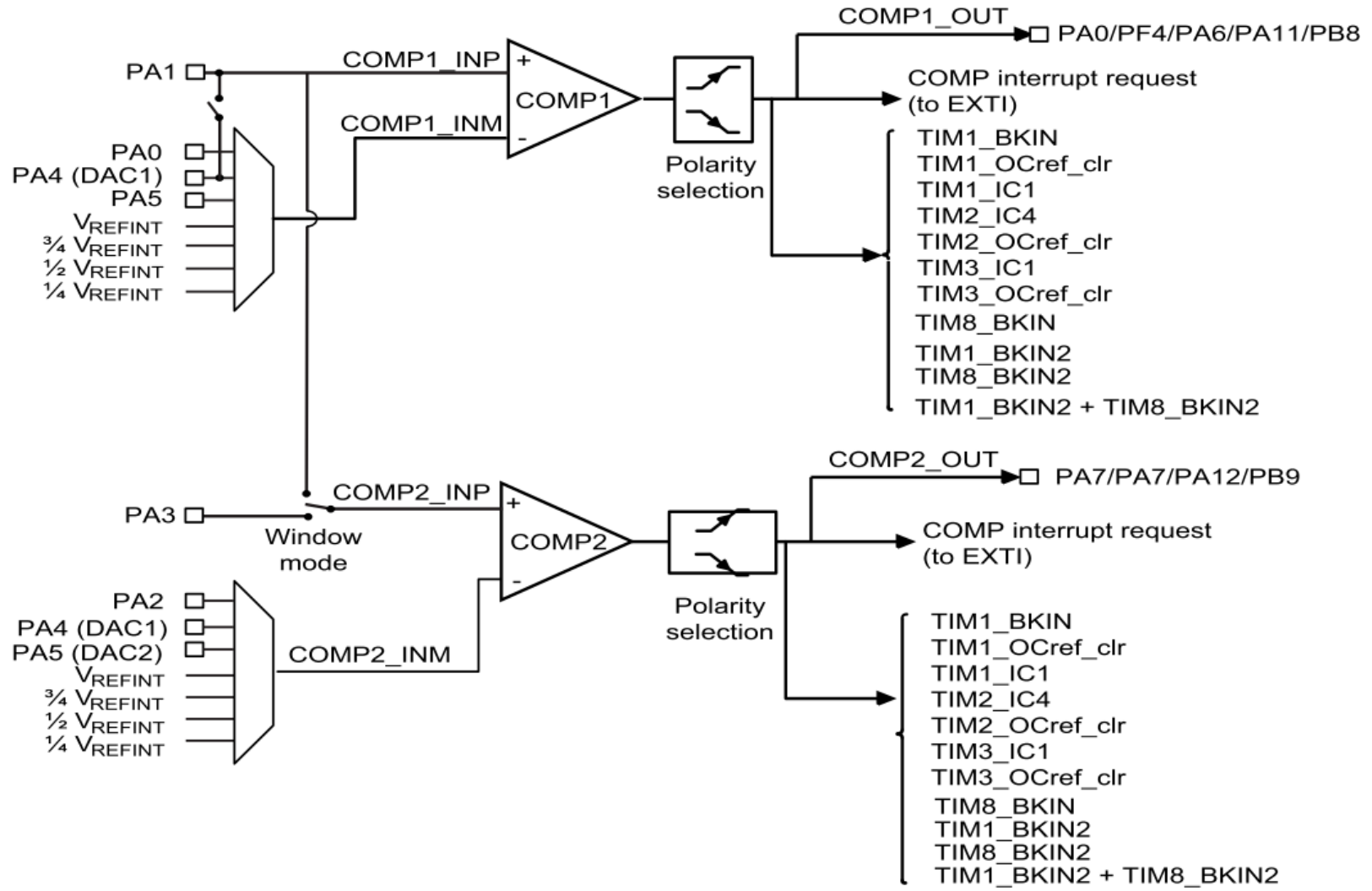
COMP features (2/2)

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- **Comparator characteristics at a glance**
 - Full operating voltage range $2V < VDDA < 3.6V$
 - Propagation time vs consumption
 - High speed / full power
 - Medium speed / medium power
 - Low speed / Low power
 - Very low speed / Ultra-low power
 - Input offset: $\pm 4mV$ typ, $\pm 20mV$ max
 - Programmable hysteresis: 0, 8, 15, 31 mV
- **Fully asynchronous operation**
 - Comparators working in STOP mode
 - No clock related propagation delay
- **Functional safety (Class B)**
 - The comparator configuration can be locked with a write-once bit

Block diagram for STM32F30x

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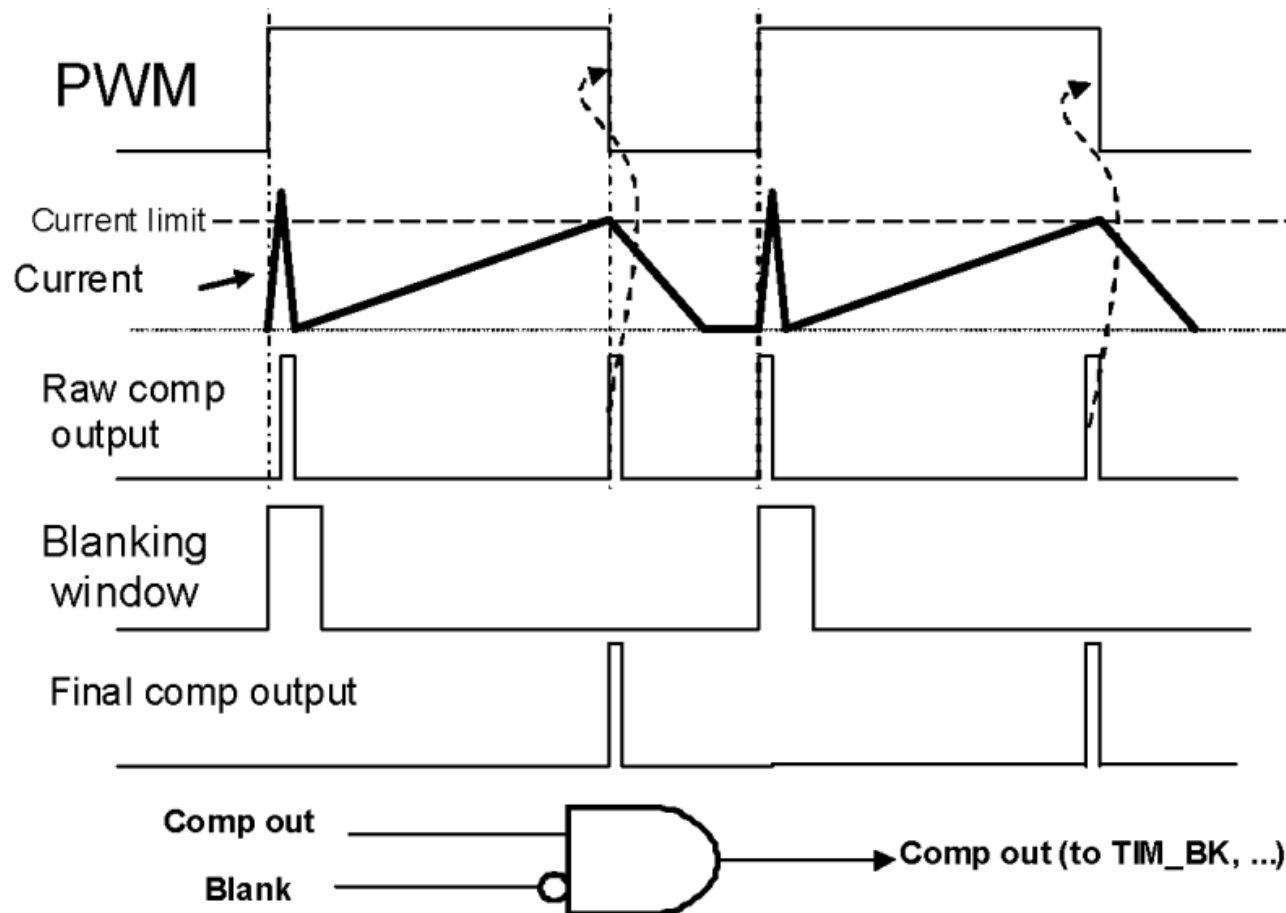
BKIN: PWM's "Emergency stop" input

OCRefClear: PWM clear for cycle-by-cycle current controller

Blanking function

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- Purpose: prevent the current regulation to trip upon short current spikes at the beginning of the PWM period (typically the recovery current in power switches anti parallel diodes).



- How many options are for internal threshold setting if DAC is used by another task?
- Can the threshold go from 0 to VDDA ?
- How can the lock bit be reset once activated ?

Operational Amplifier

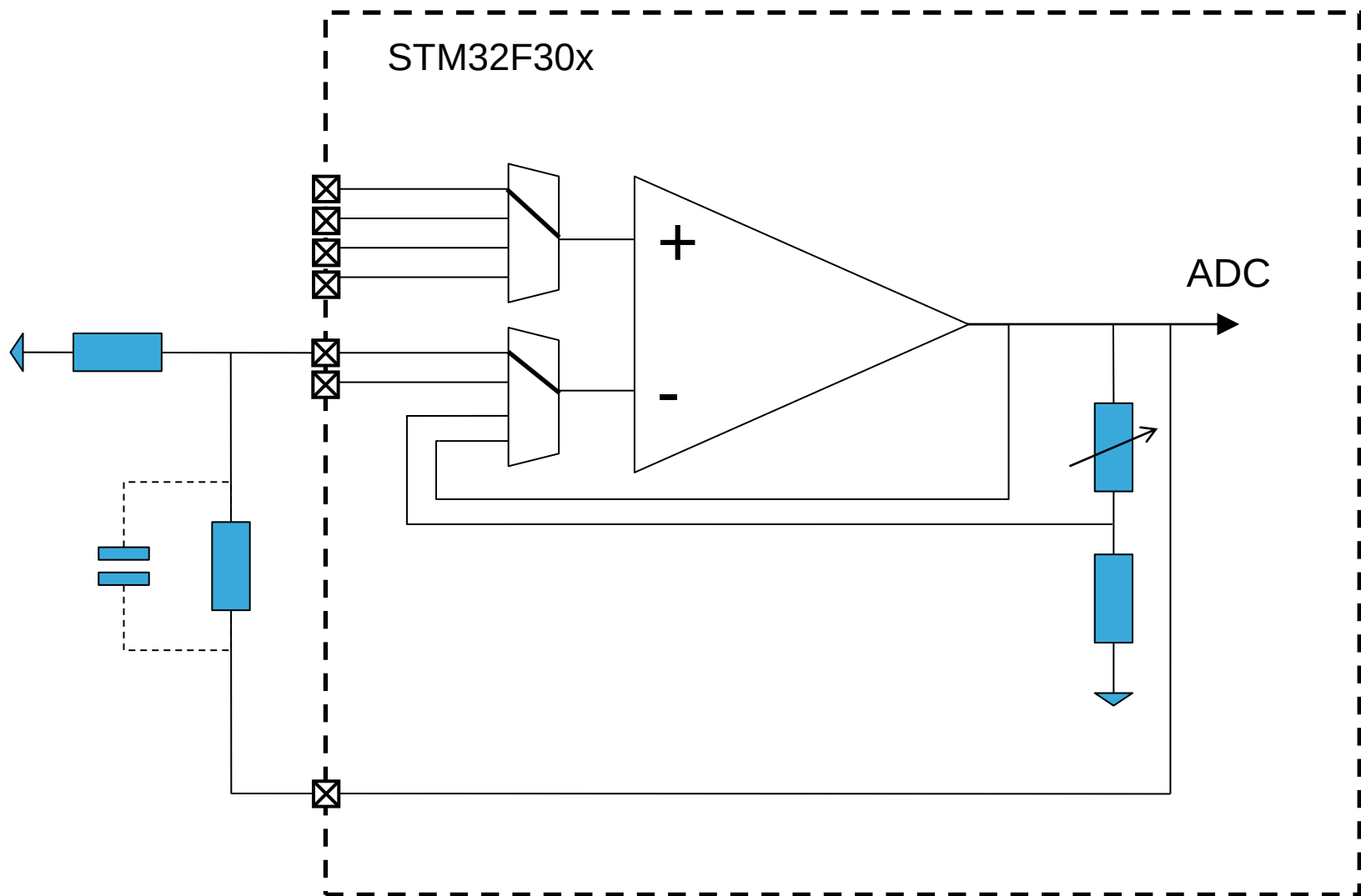


- Up to 4 operational amplifiers
- Rail to Rail input/output
- Low Offset voltage
- Access to all terminals
- Input multiplexer on inverting and non inverting inputs
- Input multiplexer can be triggered by a timer and synchronized with a PWM signal.
- 4 operating modes:
 - Standalone mode: External gain setting
 - Follower mode
 - PGA mode: internal gain setting (x2, x4, x8, x16)
 - PGA mode: internal gain setting (x2, x4, x8, x16) with inverting input used for filtering.

- Operating conditions
 - $2.4V < VDDA < 3.6V$
 - $-40^{\circ}C < Temp < 105^{\circ}C$
- Input stage
 - Input: rail to rail
 - Offset: 10mV max
 - $I_{bias} < +/-1\mu A$ max (mostly I/O leakage)
- Output stage
 - Output: rail to rail
 - $I_{load} < 500\mu A$ (sink and source)
 - Capacitive load $< 50pF$ (stable when connected internally on ADC input)
 - $GNDA + 100mV < V_{out} < VDDA - 100mV$ (Max)
- Speed
 - GBW: 8MHz
 - Slew rate $4.5V/\mu s$
 - unity gain stable

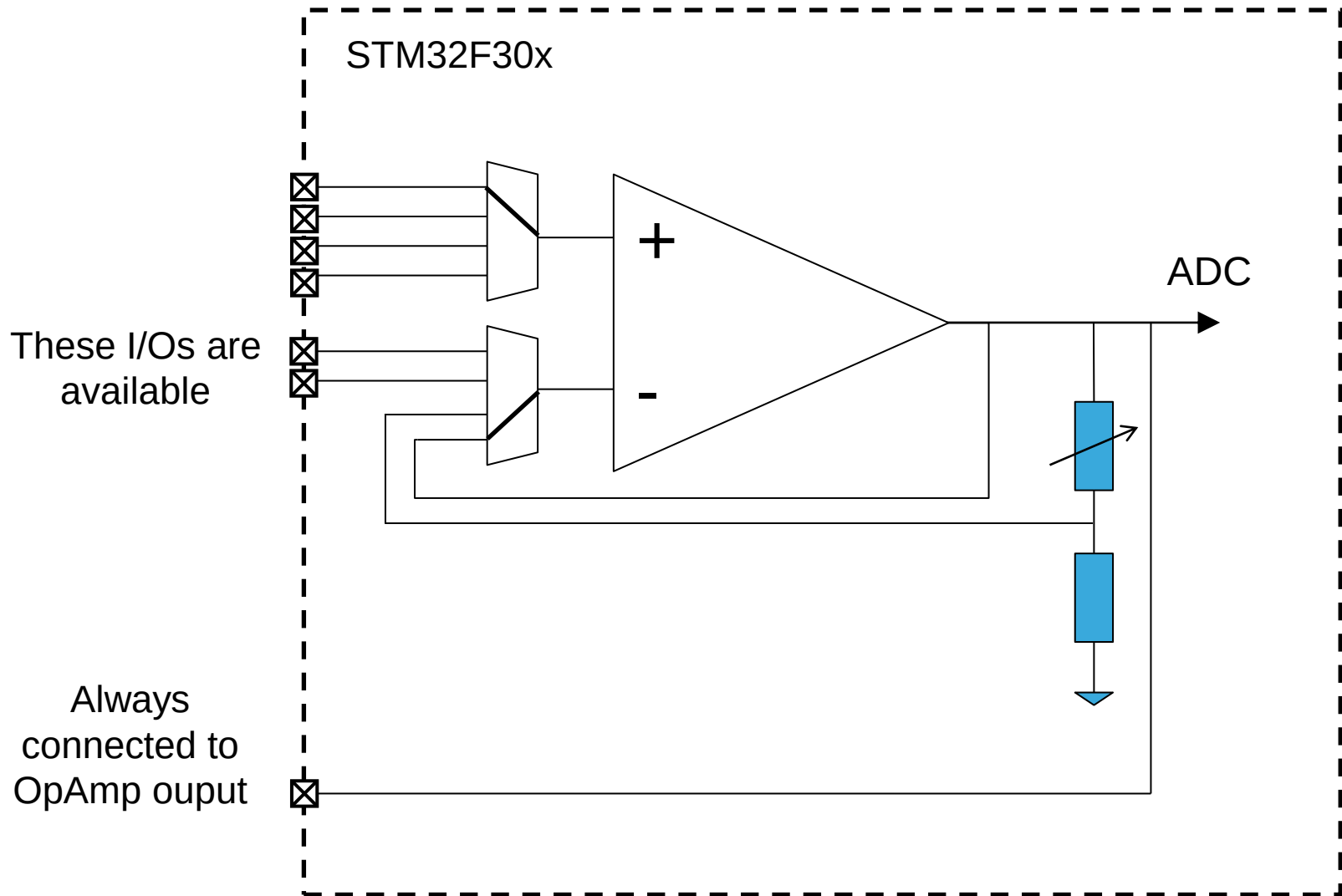
Standalone mode, External Gain setting

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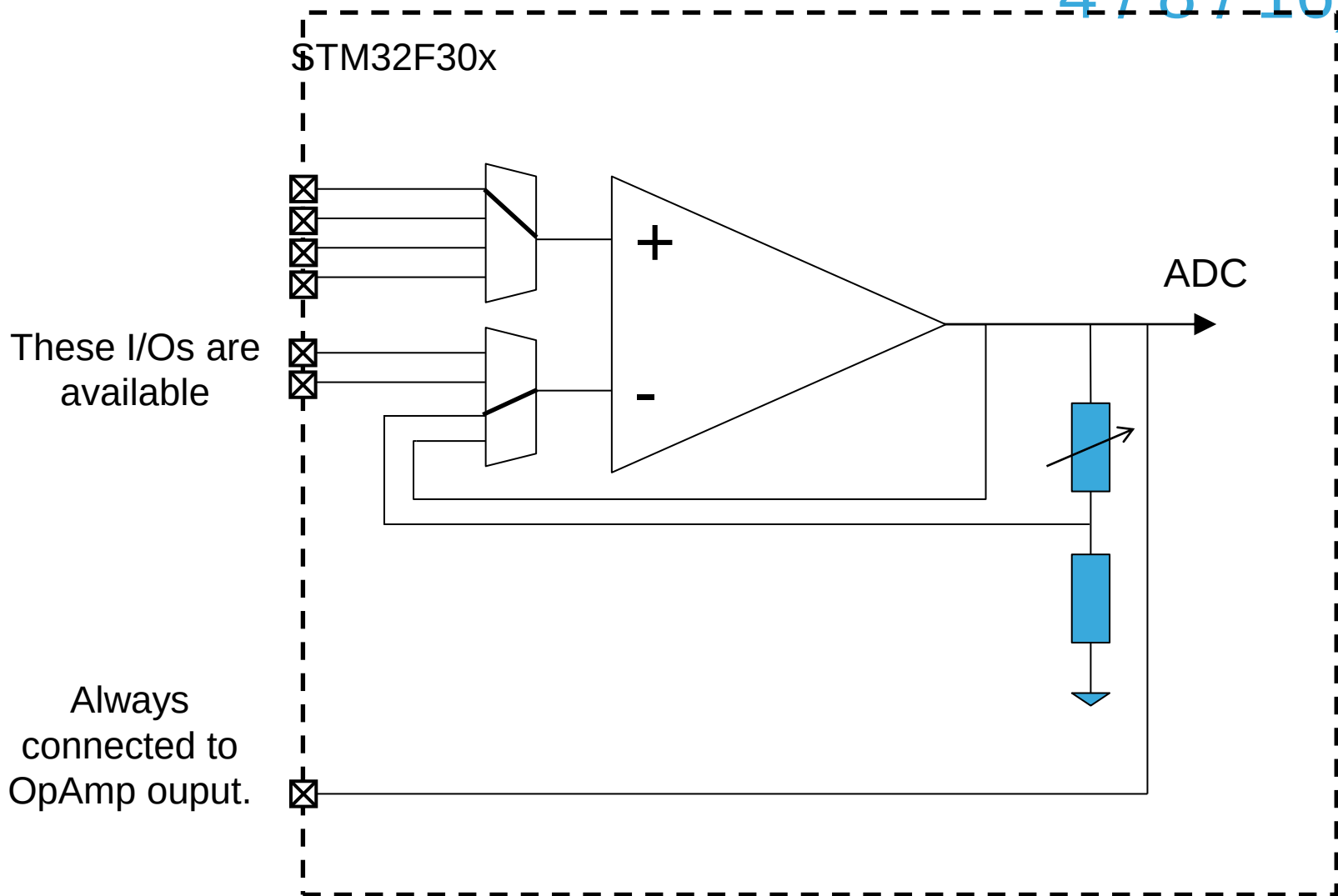
Follower mode

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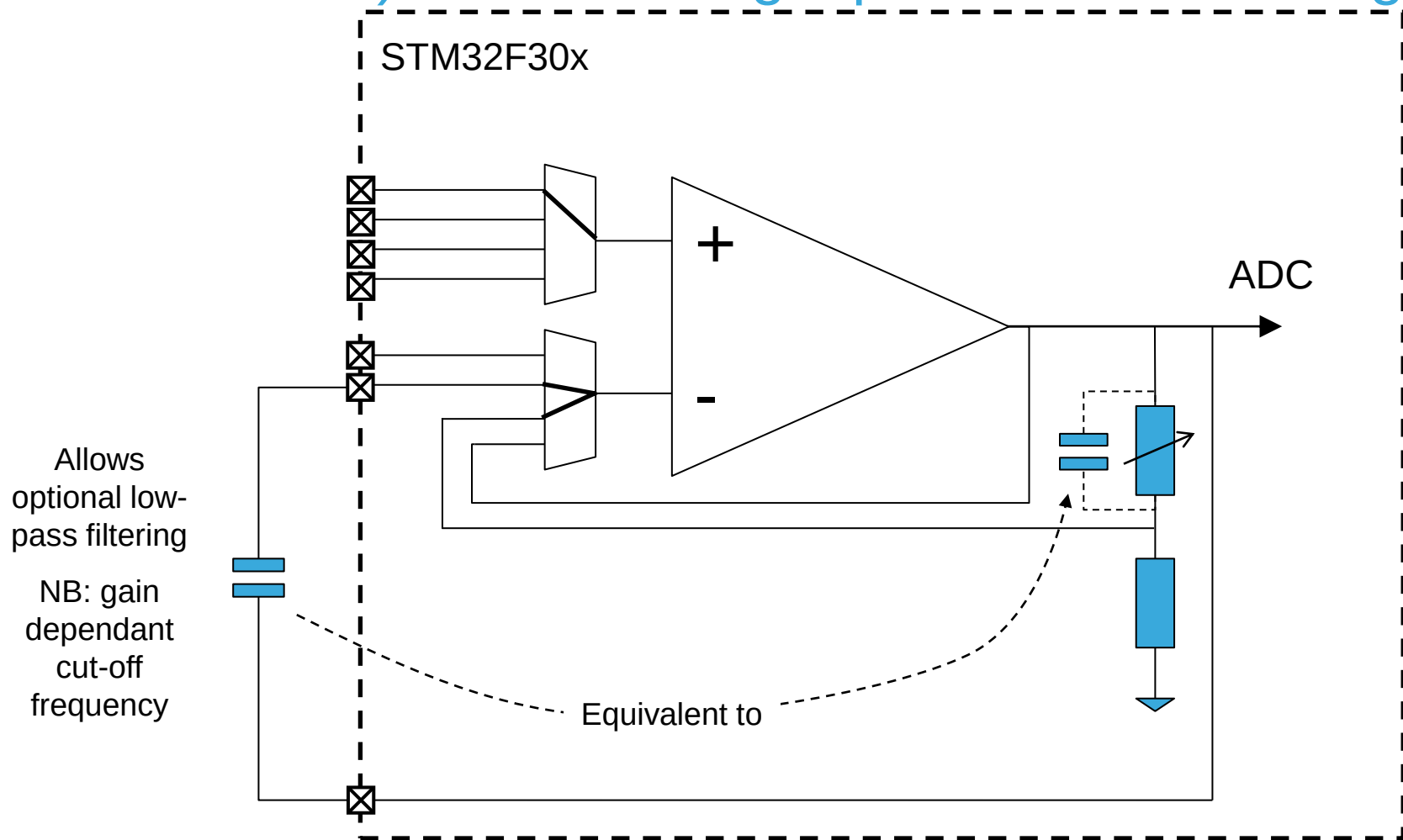
PGA Mode, Internal Gain setting (Gain = 2 / 4 / 8 / 16)

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PGA Mode, Internal Gain setting (Gain = 2 / 4 / 8 / 16) with Inverting input used for filtering.

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Timer Controlled Multiplexer mode (1/2)

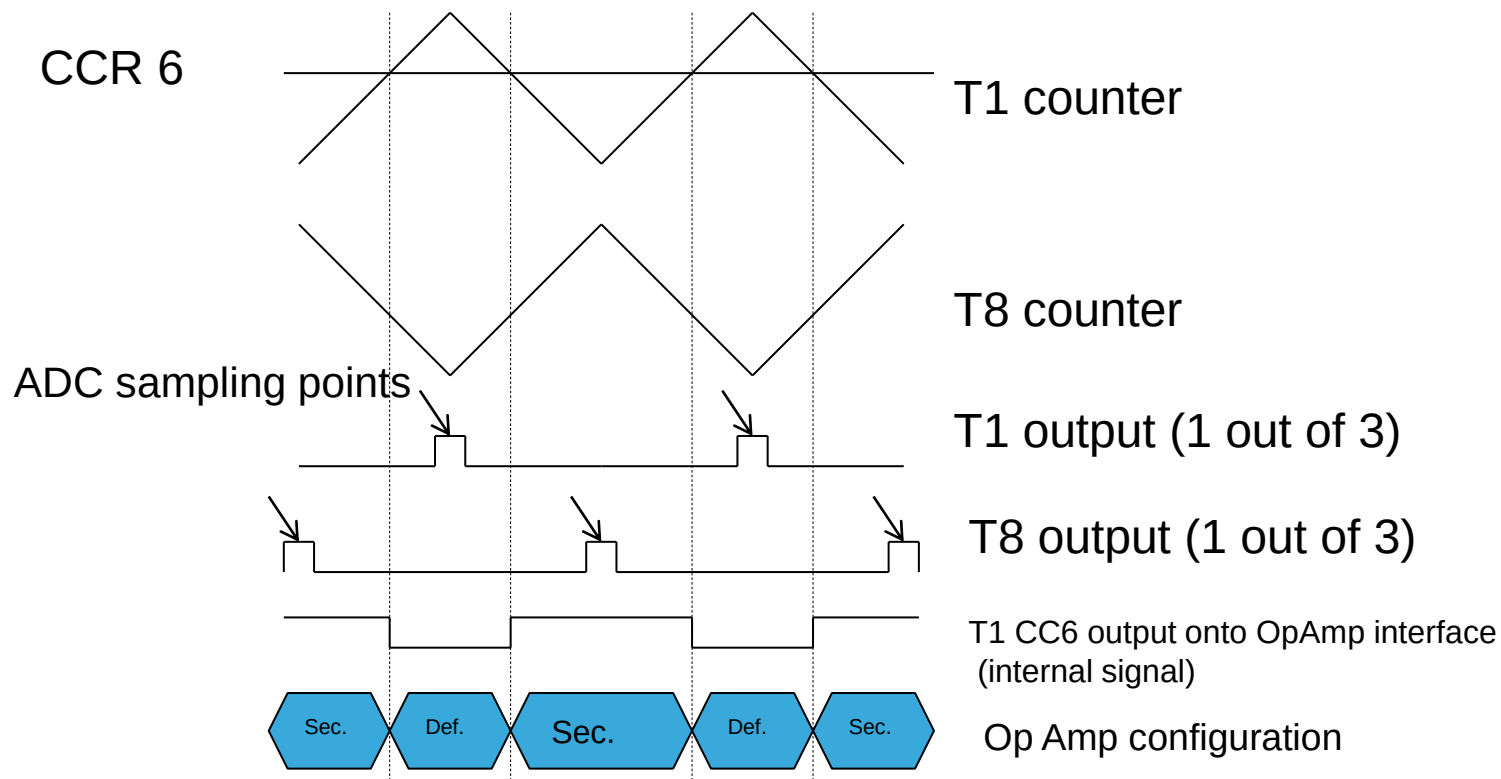
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- This mode allows switching automatically from one inverting (or non inverting) input to another inverting (or non inverting) input.
 - Benefit: useful in dual motor control with a need to measure the currents on the 3 phases on a first motor and then on the second motor.
- The automatic switch is triggered by TIM1 CC6 output arriving on the OPAMP input multiplexers.
- The Timer Controlled Multiplexer mode is enabled by setting TCM_EN bit.
- If TCM_EN bit is set, inverting and non inverting input selection is done using VPS_SEL and VMS_SEL bits.
- If TCM_EN bit is reset, inverting and non inverting input selection is done using VP_SEL and VM_SEL bits.

Timer Controlled Multiplexer mode (2/2)

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- It is possible to do the trimming of every opamp offset.
- At startup, trimmed offset values are initialized with the preset 'factory' trimming value
- The user can switch from the 'factory' values to the 'user' trimmed values using the USER_TRIM bit in the OPAMP control register.
- The offset of each operational amplifier can be trimmed by programming the TRIMOFFSETN and TRIMOFFSETP bits in the OPAMP control register.

- How many operational amplifiers are there in the STM32F30x microcontroller ?

- How many OPAMP operating modes are there in the STM32F30x?

- What is the benefit of Timer controlled multiplexed mode?

F37x COMP vs F30x COMP

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- 2 comparators (7 in STM32F30x)
- A single register manages both comparators (in STM32F30x: one register per comparator).
- No mux on the non inverting input
- No blanking feature



Thank you !

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